1 Problem statement

In the following, a monoid is an integral, saturated, sharp, commutative, unital semigroup. Monoids are written additively. A monoid M induces a translation invariant partial order on its associated group M^{gp} such that $x \geq 0$ if and only if $x \in M$. A valuative monoid is a monoid in which the associated partial order is a total order.

If M is a monoid, $x \in M^{\mathrm{gp}}$, and $y \in M$, we write $x \prec y$ to mean that there are integers m and n such that $my \leq x \leq ny$.

A monoid homomorphism $f : M \to N$ is called *infinitesimal* if the following condition holds:

- If there are $x \in M^{\mathrm{gp}}$ and $y \in M$ such that $f(x) \in N$ and $x \prec y$ and $f(y) \prec f(x)$ then $x \in M$.

If $P \xrightarrow{f} Q \xrightarrow{g} N$ are monoid homomorphisms, we say that f is *birhizomic* at g if it is of finite presentation in the category of (integral, saturated, sharp) monoids and, for every factorization of gf through monoid homomorphisms $P \xrightarrow{h} M \xrightarrow{p} N$ with p infinitesimal, there is a unique monoid homomorphism $u : Q \to M$ with $pu = g$ and $uf = h$.

A family of morphisms $P \to Q_i$ is called a birhizomic cover if, for every valuative monoid V and every homomorphism $v : P \to V$, there is a factorization of v through some $Q_i \to V$ such that $P \to Q_i$ is birhizomic at $Q_i \to V$.

Let P be a monoid. Let A_P be the category of monoid homomorphisms $P \to Q$. The birhizomic covers form a Grothendieck topology on A_P^{op}.

Let P be a monoid. For any x in P, there is an initial monoid homomorphism $P \to P[-x]^\sharp$ that sends x to 0. Let B_P be the full subcategory of A_P consisting of homomorphisms $P \to P[-x]^\sharp$, for $x \in P$.

We have given A_P^{op} the birhizomic topology and B_P^{op} the trivial topology in which all presheaves are sheaves. The inclusion $B_P^{\mathrm{op}} \subset A_P^{\mathrm{op}}$ is left exact. By left Kan extension and sheafification, it extends to a functor $\pi^{-1} : \mathrm{Sh}(B_P) \to \mathrm{Sh}(A_P)$ on the categories of sheaves. Let π_* be the right adjoint of π^{-1}.

1.0.1 Question *Is it the case that, for all presheaves of abelian groups F on B_P^{op}, the morphism*

$$F \to \mathrm{R}\pi_*\pi^{-1}F$$

is a quasi-isomorphism?